Exercises

Exercise 6: Graphics for communication

Read Chapter 5 to help you complete the questions in this exercise.

Before the session. This exercise uses the `ggplot2` package, which doesn't come with R, so please install it before you arrive. Run `install.packages("ggplot2")` in the console. You only need to do this once, but you will need to use `library(ggplot2)` at the beginning of any script that uses it.

Everything you plotted in Exercise 5 was for graphical data exploration (GDE). Exploratory plots are quick, sometimes a bit ugly, and their only job is to show you what your data looks like. It doesn't matter that the axis label says `cardiac$tchol` because you're the only person who'll ever see it. This exercise is about the other kind of plot, the one that goes on a poster or in a report, where the reader isn't you, has about fifteen seconds, and won't ask you what the axis means.

We'll also change graphics package for this exercise. Base R, which you used in Exercise 5, is great for GDE. For a figure somebody else has to read, `ggplot2` is (arguably) the better tool. It builds a plot up in layers, it works out the legend for you, and it lets you say what you want rather than how to draw it, which leaves you free to think about the decisions this exercise is really about. Having said that, this doesn't mean the `ggplot2` package is inherently better than base R, it's just a different tool for the job.

We will also take this as an opportunity to switch datasets (just in case you are getting bored with the *cardiacdata* dataset!). In this exercise we'll use data from the Scottish Public Health Observatory.

1. As in previous exercises, either create a new R script or continue with your previous R script in your RStudio Project. Again, make sure you include any metadata you feel is appropriate (title, description of task, date of creation etc) and don't forget to comment out your metadata with a `#` at the beginning of the line.

2. Download the data file *'scotpho_alcohol_admissions.txt'* from the **Data** link and save it to your `data` directory. This one is already a tab delimited text file, so there is no need to convert it. Import it with `read.table()` (just as you did in **Exercise 3, Q5**), remembering to write out your `header`, `sep` and `stringsAsFactors` arguments, and assign it to a variable called `scotpho`. Then make `ggplot2` available for the rest of the session with `library(ggplot2)`.

3. These data are an extract from the ScotPHO online profiles tool. The indicator is *alcohol-related hospital admissions*, and the `measure` column is an age-sex standardised rate per 100,000 population: in other words, how many people per 100,000 were admitted to hospital for a reason related to alcohol, adjusted so that areas with different age and sex profiles can be compared fairly. The extract covers each of the 32 Scottish

council areas plus Scotland as a whole (`area_name`, `area_code`, `area_type`), for each year from 2010 to 2019 (`year`). Because the rate is an estimate rather than a count, it comes with a 95% confidence interval (`lower_confidence_interval`, `upper_confidence_interval`), which gives you a sense of how precise the estimate is (we'll come back to these in Q10 if you have time).

4. Before plotting anything, get your bearings, exactly as you did in **Exercise 3, Q6 and Q8**. Use `str()` and `summary()` on `scotpho`. How many rows are there? Use `table()` (you met this in **Exercise 4, Q5**) on `area_type` to see how the rows split between council areas and Scotland. Then check something that might matter before you draw a line through points: does every area have a value for every year? (hint: `table()` on `area_name` will tell you how many rows each area has).

5. Right, let's plot something. Start with the simplest possible communication plot: how have alcohol-related admissions across Scotland as a whole changed over the ten years? Extract just the rows where `area_name` is "Scotland", using the square bracket notation from **Exercise 3, Q10**, and put them in a dataframe called `scot`.

Now for your first ggplot. Every ggplot is built out of the same three pieces (Section 5.2.1 of the book walks through them slowly):

- the **data**, the dataframe you're plotting
- the **aesthetics**, written inside `aes()`, which say which column of that dataframe goes on which part of the plot. Here it's `year` along the x axis and `measure` up the y axis
- a **geom**, which is the thing actually drawn. `geom_line()` draws a line and `geom_point()` draws points, and you can have both

You join them together with `+`, one layer per line. Here is the shape of it, with the parts you need to change in capitals:

```
ggplot(data = YOUR_DATAFRAME, aes(x = YOUR_X_COLUMN, y = YOUR_Y_COLUMN)) +
  geom_line() +
  geom_point()
```

Fill it in and run it. Then look at what ggplot has given you by default, and list everything a stranger would not understand about it.

6. OK, a single line rarely tells the whole story. Scotland's average hides a lot of variation between council areas, so let's add two of them, "Glasgow City", which has the highest rate in Scotland, and "Aberdeenshire", which has one of the lowest.

This is where ggplot starts to earn its keep. In base R you would pull each area into its own dataframe and draw three separate lines, keeping the colours and the legend in step by hand. ggplot works the other way round. You give it one dataframe holding all three areas, tell it which column says which area a row belongs to, and it draws one line per area and builds the legend for you.

a) First we need to get those three areas into a single dataframe. You could do it with three conditions joined by `|`, the 'or' operator, but there's a tidier way. The `%in%` operator takes a value on its left and a vector on its right and asks whether that value appears anywhere in the vector. Hand it a whole column and it does that for every row in turn, giving you back a `TRUE` or a `FALSE` for each one, which is exactly what the square brackets want. It's the same job you did with `==` and `&` in **Exercise 3, Q10**, just against a list of possibilities rather than a single one.

So `scotpho$area_name %in% areas` gives you 330 `TRUE`s and `FALSE`s, one per row of `scotpho`. Make a vector called `areas` holding the three area names, use it to extract those rows into a dataframe called `three_areas`, and check with `nrow()` that you have the 30 rows you expect. The nice thing about writing it this way is that it doesn't get any longer if you decide you want six areas instead of three. Don't worry if you don't get this the first time round, take a peek at the solutions which will hopefully make it a bit clearer!

b) Now let's plot it, with the same three pieces as Q5 and one addition. Inside `aes()`, alongside your `x` and `y`, add `colour = area_name`. That single argument is what tells ggplot the rows belong to three different areas, and it does the rest for you, a line each and a legend down the side (see Section 5.2.4 for more on linking colour to a variable). What do the three lines tell you that the Scotland line on its own didn't?

7. Now, let's think about those colours. ggplot's defaults are not too bad, but they're not chosen with anybody's eyesight in mind. Roughly 1 in 12 men has some form of colour vision deficiency, most commonly difficulty telling red from green, and red and blue aren't a safe pairing either.

a) Let's redraw the plot with a colour-blind friendly palette. Base R has one built in, and `palette.colors(3, palette = "Okabe-Ito")` will give you three colours from it. Assign these to `cols`, then hand them over to ggplot by adding a `scale_colour_manual(values = cols)` layer onto the end of your plot. A scale layer is how you overrule any of the decisions ggplot has made for you (see Section 5.2.4 for more).

b) Now let's test it. Posters get photocopied and printed in black and white, so it's worth knowing whether your figure still works with no colour at all. Draw the plot again, this time swapping those colours for their greyscale equivalents, `grey(c(0, 0.64, 0.62))`, and see whether you can still tell the three lines apart. If you can't, what could you change other than the colour?

c) So let's fix it. Colour on its own is rarely enough, so it's a good idea to give your reader a second cue as well. Add `linetype = area_name` inside `aes()`, next to the `colour = area_name` you already have, and each area gets its own style of line as well as its own colour. Notice that ggplot works out that the two are describing the same thing and gives you a single legend rather than two.

8. Next, let's sort out the text. Every plot needs axis labels a non-specialist can read with units on them, a title that says what the reader is looking at, and a note saying where the data came from. Write the labels for someone who has never heard the phrase 'age-sex standardised'. Your reader is a public health colleague or a member of the public, not a data scientist. We'll revisit this again in Q9.

In ggplot all of the words live in a single `labs()` layer, which takes `x`, `y`, `title` and `caption` arguments (see Section 5.2.6). The caption is where the source note goes. While you're there, adding `colour = NULL` and `linetype = NULL` to the same `labs()` will get rid of the 'area_name' heading above the legend, which is another column name your reader doesn't need to see.

There are three more things to sort out, and each one is another layer.

a) The x axis is still labelled with half years, which is a bit odd. Add `scale_x_continuous(breaks = seq(2010, 2019, by = 2))` to put the breaks on whole years instead (see Section 5.3.10).

b) Now have a think about the y axis. Should it start at zero? There's no single right answer here, but you should be able to justify whichever you go for. If you decide it should, `scale_y_continuous(limits = c(0, 1600))` will do it (see Section 5.3.2).

c) Finally, the overall look of the plot. ggplot's grey panel is fine on screen but can come out a bit muddy in print. A **theme** controls everything about the plot that isn't your data, and there are several ready-made ones to choose from (see Section 5.2.5). Add `theme_minimal()` and see what you think. If you'd rather the legend sat underneath the plot than beside it, `theme(legend.position = "bottom")` will move it (see Section 5.3.6). Just add it after `theme_minimal()`, otherwise it gets overwritten.

9. Last of all, let's get the plot out of R at a size and resolution that's fit for a poster. The screen defaults aren't much use here, and a plot that looks fine in the RStudio plot pane might be a blurry mess blown up to A1.

This is a bit easier if you give the plot a name first. Assign the whole thing from Q8 to an object called `admissions_plot`, exactly as you would assign anything else in R (there's an example of this in Section 5.2.3). Don't worry when nothing appears, that's normal. You've stored the plot rather than drawn it, so type `admissions_plot` in the console to see it.

Now save it with `ggsave()`, which is ggplot's version of the three step business you met in **Exercise 5, Q12**. There's no device to close this time, because `ggsave()` opens and closes it for you, and it works out which format you want from the file extension you give it (see the end of Section 5.2.6).

Save your plot twice into the `output` directory you created in Exercise 1, once as `ex6_admissions.pdf` and once as `ex6_admissions.png`. Use `width = 10` and `height = 5.625`, which gives you a 16:9 shape and is what most poster and slide templates expect, and for the png also set `dpi = 300`, which is the usual minimum for print. Open both files and have a look at them at full size before you believe them.

Hang on to that png. You'll put it into a report in Exercise 7 and then into a repository in Exercise 9, so it needs to still be there in a fortnight.

10. (Optional) This last one is for those of you who have a bit of time left, or who are just curious. Each value on your plot is a point estimate, a single best 'representation' of the admission rate for that area in that year. On its own it doesn't tell your reader anything about how precise it is. Remember back in Q3 there were two extra columns in the data, `lower_confidence_interval` and `upper_confidence_interval`? Let's use them to show that uncertainty on the plot.

Start with the code for your plot from Q9 and add two layers to it. A `geom_point()` layer puts a point on each year (you used this back in Q5), and a `geom_errorbar()` layer draws the confidence interval around it. `geom_errorbar()` needs to know where each bar starts and finishes, so give it its own `aes()` with `ymin` set to the lower confidence interval column and `ymax` set to the upper one. Add `width = 0.2` as well, otherwise the caps on the ends of each bar are almost a year wide, and `linetype = "solid"`, otherwise the bars pick up the dashed and dotted lines from the main `aes()`. Notice that these two go outside the `aes()`, because they're fixed values rather than columns in your data (Section 5.2.1 explains the difference, and why an `aes()` inside a geom only applies to that geom). ggplot draws the layers in the order you add them, so put `geom_point()` after `geom_errorbar()` to get the points on top of the bars (see Section 5.3.3). And don't forget to update your caption so the reader knows what the bars are. If you get stuck, type `?geom_errorbar` in the console to open the help page (the examples right at the bottom are the most useful bit).

Keep an eye on your y axis limits from Q8b. The upper confidence interval is always a bit higher than the rate itself, and if a bar doesn't fit inside your limits ggplot just leaves off the part that doesn't fit, usually without any warning, so it's easy to miss (Section 5.3.2 has more on what happens to data outside your axis limits). `max(three_areas$upper_confidence_interval)` will tell you how high your upper limit needs to go.

Once you've got it working, have a look at how wide the bars are. Why do you think the bars for Scotland are so much narrower than the ones for Glasgow City and Aberdeenshire?

If you'd like to keep this version, save it in the same way as Q9. Assign it to an object called `admissions_ci_plot` and save it with `ggsave()` as `ex6_admissions_ci.png` in your `output` directory. Make sure you give it this new name, otherwise you'll overwrite your `ex6_admissions.png`, and you'll need that one in Exercise 7.

End of Exercise 6